

ARUNKUMAR S

☎ + (91) 9585433503 ✉ arunaksv1@gmail.com [in linkedin.com/in/Arunkumar S](https://www.linkedin.com/in/Arunkumar S) github.com/arunaksv

Professional Summary

Innovative Software Engineer with around 2.5 years of experience in developing and optimizing high-performance applications. Specialized in C/C++ with a strong background in Software Development as well as embedded systems and proficiency in Linux environments. Skilled in all phases of the software development lifecycle[Agile], including requirement gathering, design, coding, and testing. Adept at applying Object-Oriented Programming principles to create robust, scalable solutions and experienced in producing reusable code to achieve project goals.

Work Experience

Senior Software Engineer

Bangalore, Karnataka

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July 2022 - present

- **Application Development:** Development of high-performance applications using C++, focusing on optimizing functionality, ensuring efficiency, reliability, and tailoring solutions to meet specific user needs.
- **Stack Development (Protobuf & OSI Layers):** Created a stack incorporating Protocol Buffers for efficient data serialization and OSI layers for encoding and decoding messages, ensuring robust data handling and communication.
- **Protocol Integration(IPC):** Implemented UDP and VSomeIP protocol support for inter-process communication, integrating it within the stack library to enhance system communication and connectivity.
- **Multithreading & Memory Management:** Utilized multithreading for concurrent data processing and employed smart pointers for effective memory management, ensuring stability and reliability of the system.
- **Interface Development (gRPC):** Designed and implemented a gRPC-based communication interface for the Central ECU, enabling efficient and seamless data exchange across distributed automotive systems.
- **Development Environment:** Worked extensively in Linux environments, using Visual Studio Code (IDE), command-line tools, and development utilities for efficient coding, debugging, and deployment.

Skill Summary

Programming Languages: C , C++ 11/14/17, Bash Scripting, Socket Programming.

Domain Specific Skills: CMake, STL, V2X, gRPC, Connectivity, Linux, SELinux, System/Software Architectural Design, Agile/Scrum, MQTT, GoogleTest(Unit testing), TCP/IP, UDP, VSome/IP, BOOST Library, GNU Debugger(GDB).

Tools: Git, Jira, Confluence, UML, EB/Drawio, Static Analysis, lcov, gcov, Valgrind, PuTTY, Vim, Visual Studio Code, Postman, Eclipse, CentOS, ubuntu, Windows, MakeFile, SQLite, Wireshark, Enterprise Architect(EA), Virtual Machine, Microsoft 365.

Projects

DriveSafe Dashboard!

- Created a dashboard alerts to display critical safety notifications to drivers regarding seat belt alerts, overspeeding detection, and brake alerts to enhance road safety.
- Developed APIs for real-time monitoring of seat belt status, vehicle speed, and brake performance, ensuring effective communication between components.
- Analyzed simulation results to identify potential issues and optimize alert mechanisms, ensuring accurate and timely notifications to drivers.
- Collaborated with cross-functional teams to refine the user interface based on results from simulated testing, enhancing usability and driver interaction.

SELinux - ControlHub!

- Designed and implemented custom SELinux policies to enforce Mandatory Access Controls(MAC), enhancing security for various applications and services in a Linux environment.
- Developed APIs to maintain logs and manage access control, facilitating real-time monitoring and auditing of security events within the system.
- Integrated SELinux with IAM, conducted audits to optimize policies based on logs, and used audit2allow to resolve violations, streamlining authentication and authorization.

Education

Kumaraguru College of Technology

Aug 2018 – May 2022

Bachelor of Engineering in Electronics and Instrumentation CGPA: 8.9

Coimbatore, Tamilnadu

Vivek Vidyalaya Matric Higher Secondary School

Aug 2017 – May 2018

Class XII Percentage: 92

Coimbatore, Tamilnadu

Certifications

- OOPs Concepts in C++ | [Certificate](#)
- Advanced C++ | [Certificate](#)
- C++ Essential Training | [Certificate](#)
- Software Design: Modeling with UML | [Certificate](#)
- C++ Development: Advanced Concepts, Lambda Expressions, and Best Practices | [Certificate](#)